

CPD Project Final Report

CS 396 Causal Inference

1. Group Members

Alina Chen, Yulan Guo, Qinyan Li, Haohan Shi

2. Code and Documentation

`cpd_models.ipnyb`: contains our final backdoor estimator, the bootstrapped confidence intervals, our doubleML model using random forest, and our preliminary model with final finding as the outcome.

`cpd_synthetic_data.ipynb`: has the code we used to generate our synthetic data, and compares the expected and observed causal effect with a naive estimate at sample sizes of 1,000, 10,000 and 100,000.

3. Estimation Implementation

Our estimation code can be found in cell 27 of the `cpd_models` notebook, and the results can be found in the output of cell 29. The bootstrapped estimate code is found in cell 30.

We utilized backdoor estimation to isolate the causal effect of rank on `penalty_code` for sustained complaints, while controlling for confounding effects from race (C1), gender (C2), age (C3), and allegation category (C4).

Our counterfactual function is:

$$E[Y^a] = \sum_Z p(Z) E[Y|A = a, Z] \text{ where } Z = \{C_1, C_2, C_3, C_4\}$$

We made the typical assumptions necessary for backdoor estimation, including consistency, conditional exchangeability, and conditional independence of the rank and the counterfactual outcome given our confounders.

Since we're building one model for each rank, the four model parameters are the coefficients of the four confounders we're including in our model (race, gender, age, and allegation category). Since we're using a logistic regression model, these parameters are used within each model to predict the probability of the outcome variable ('penalized') for each sample. For example, the model for rank 2 ('police officer') has parameters [-0.170, 0.295, 0.539, -0.151], while the model 8 representing the combined highest ranks has parameters [-2.48, 0.156, -0.214, -0.224].

By building a separate model for each rank, we're able to maximize flexibility and minimize unnecessary assumptions. The varying model parameters indicate that our confounders have different effects depending on the rank of the sample, which shows that our model setup is able to handle any nonlinear relationships between the confounders, rank and outcome.

4. Changes Since the Update

Since the update, our main changes have been to add complexity to the model with additional confounders, create bootstrap confidence intervals to determine the statistical significance of our estimates, and experiment with more complex models.

4.1 Additional Confounders

We refined our DAG from the update, switching out the appointment date for age, which we felt like could encapsulate most of the confounding from the length of time that the officer had served while also accounting for other effects like perceived authority or expertise. In our backdoor estimator, we added in the allegation category and age as covariates, and converted the allegation category into a binary variable where 1 represented complaints regarding violent incidents, and 0 was nonviolent. We continued to only investigate sustained complaints in our model.

The code for our updated model can be found in cell 27 of the `cpd_models` notebook, and various preprocessing code for each of the variables is located above.

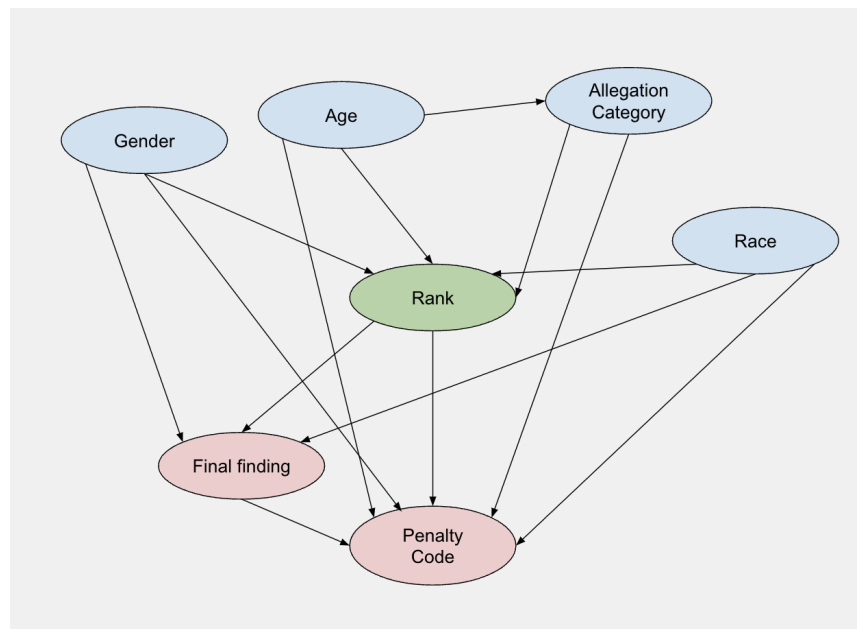


Figure 1: Final DAG

4.2 Bootstrap Confidence Intervals

Although our estimation results from both the update and the final model showed a very promising negative correlation between rank and penalization rate, we wanted to verify the statistical significance of our results with bootstrapping. In cell 27, generated 95% confidence intervals on each rank estimate, and found that there were no overlaps in the intervals, indicating a statistically significant difference in the treatment effects for each rank.

4.3 Complex Models

We also wanted to experiment with using more complex models such as DoubleML for our causal problem, especially since they would theoretically require less assumptions about categorical variables such as allegation category or final finding.

In cell 28, we implemented a DoubleML estimator from the econML library, and experimented with using LassoCV and RandomForest models. It was difficult to find a suitable model that could handle multiclass treatments, so for our initial model we chose to group ranks 2-5 into a “low” rank and 6-10 into the “high” rank. With this binary treatment, we found an average treatment effect of -0.0938, which indicates that if an officer is promoted, their chances of facing punishment following a sustained complaint decreases by approximately 9%.

We weren’t able to convert the model to handle multiple rank treatments, but we believe that we would have needed a combination of doWhy and DML to create a more robust model. Also, our DML model utilized gender, race, our binary variable for allegation category, age, and the amount of years the officer had served in their current rank as confounders. We wanted to add in more categorical variables such as the raw allegation category or final finding, but we weren’t able to debug errors in the model when we used one hot encoding on the variables, and adding in too many confounders significantly increased the amount of time it took the model to run and converge. With more time, we’d like to experiment more with adding in additional confounders and playing around with the complexity of the trees (model depth, n_estimators, etc).

4.4 IPW

Due to the discrepancies in the data distribution of the confounders (the sample is predominantly white and male), we experimented with implementing IPW to create weighted samples that would allow us to assess the treatment effect independence of the imbalanced confounder distribution. First, we estimated the propensity scores, which represent the probability of being in a particular rank given the confounders, using logistic regression. These propensity scores were then used to calculate the inverse probability weights for each individual. We applied these weights to our logistic regression models to adjust for confounding and obtain unbiased estimates of the treatment effect. Separate models were trained for each rank to evaluate the

effect of rank on the probability of being penalized. The code for our IPW model can be found in cell 28 and onwards in `all_models.ipynb`.

The inclusion of additional confounding variables generates more NaN values, making more unusable rows for each rank. We remove the ranks with sample size smaller than 1000, resulting in only 5 ranks in increasing order in the following table.

Rank	$E[Y^a]$
Police Officer	0.8785958725586697
Field Training Officer	0.735149241416497
Detective	0.8327177914868866
Sergeant	0.8173733870227128
Lieutenant	0.5782803879424994

We didn't spend as much time tweaking this model as we wanted, but from the initial results we can see that there is still a generally negative correlation between the rank and predictions, but it is less noticeable than in the backdoor estimator alone. This means that we might have overestimated the results with our final model, which indicates that we might want to perform sensitivity analyses to see how robust our model is against potentially unknown confounders, or we might want to continue to refine our propensity model to see if our results really do need further adjustment.

5. Interpreting your results

Before

Our changes since the update didn't change the structure of our original model, so our interpretation of the previous results remains the same.

Rank	Police Officer	Field Training Officer	Investigator	Detective	Sergeant	Lieutenant	Commander
$E[Y^a]$	0.844	0.763	0.785	0.800	0.772	0.663	0.484

We see a pretty clear overall downward trend in the probability that an officer is reprimanded for a complaint as their rank increases. The causal effect is seen most strongly at the two ends of the rankings, as a regular police officer is almost 1.7x more likely to face punishment following a complaint compared to the highest ranking commander. Therefore, there seems to be a strong indication that as officers rise through the ranks and gain more authority within their departments, they face less internal accountability for their actions.

After

In our final estimation, we decided to combine the highest three ranks, since there weren't enough sample sizes in each of the individual ranks to build a robust model. We also combined the investigator and detective groups for the same reasoning.

Rank	Police Officer	Field Training Officer	Investigator/ Detective	Sergeant	Lieutenant	Captain/Commander/ Deputy Chief
$E[Y^a]$	0.819	0.636	0.753	0.728	0.662	0.479

From these final point estimates, we can see that there is a relatively consistent decrease from the lower to higher ranks, validating our expectation that at higher levels of authority, officers face less consequences for their actions. When comparing the highest and lowest ranks, we can see that the probability of suspension decreases by 41%, which means that a police officer is 1.7x more likely to be punished compared to a high level officer. The difference between adjacent ranks is still significant: police officers are 8% more likely to face suspension compared to detectives, and sergeants are 9% more likely to be punished than lieutenant officers.

One notable counterexample to this trend is the field training officer rank, which has significantly lower levels of punishment despite not being a higher rank. We believe that this might be because field officers have significantly less interaction with the general public, and are primarily engaged in training other officers rather than traditional duties of police service. This might introduce unknown confounders for this specific rank that we were unable to account for in our model, as these alternate responsibilities might result in a different accountability system for these officers. One explanation could be that these officers are more likely to be reprimanded, rather than face suspension or separation, but we didn't consider reprimands as punishment in our model. This discrepancy could be because field training officers are held to higher standards: they have direct impact on training new recruits and act as role models for them, so maybe they're more likely to face reprimands more often to ensure exemplary behavior. They also might be more respected officers with specific expertise, which might mean they're less likely to be punished.

Our bootstrap confidence intervals provide another method to evaluate our results, and they show that the punishment rates deviate greatly based on the rank. The difference is the most stark in the lowest and highest ranks: rank 2 (police officers) had a 95% CI of [0.818, 0.822], while the highest rank had a CI of [0.432, 0.530]. With the most conservative estimates of the difference (taking the upper CI for the higher rank and lower CI of the lowest rank), a police officer is still more than 1.5x more likely to face punishment following a sustained complaint compared to a captain, commander, or deputy chief.

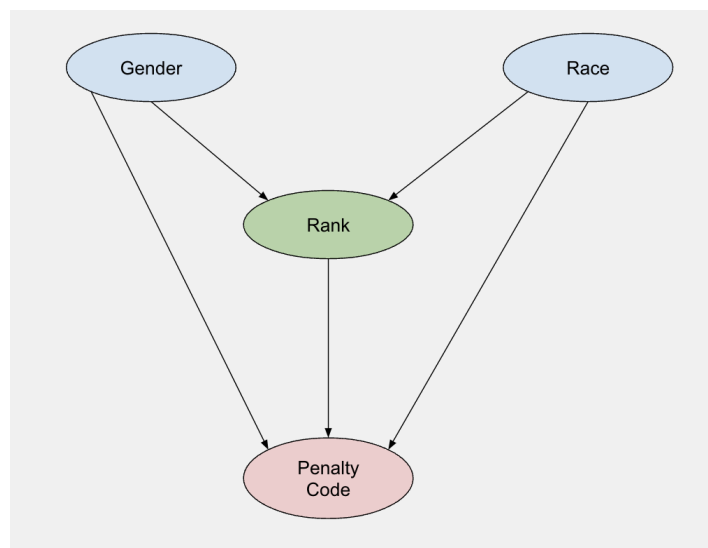
The confidence interval is rather large at the highest rank (spans around 0.1), especially compared to the intervals for lower ranks (span ~0.01). This is due to the lack of data available for the highest ranks, which makes sense because very few officers are promoted to this leadership level in their careers. Even with the comparably small sample size, our CI indicates that the difference between the ranks is still statistically significant, no matter where the true level lies within that range.

Before vs. After

Our estimates of causal effect didn't change greatly between our previous model and our final one, but the additional confounders and addition of bootstrap CIs helped us to refine our analysis and results. The lack of significant shifts with our new model may indicate that race and gender may truly be the most impactful covariates when it comes to determining punishment following a complaint, which is emblematic of an institution that has long been plagued by structural discrimination against minority groups.

6. Synthetic Data

We used the same code for our synthetic data generation as in the update, which is modeled off of the simplified DAG shown below.



For each of the confounder variables (race and gender), we used a binomial distribution to generate the values where a 1 for gender represented male, and a 1 for race represented white. We converted our treatment to binary again where 0 was the “low” rank and 1 was the “high” rank, and then created a 2 element vector of random weights between 0 and 0.5 to represent the probability of suspension for each rank, and sorted it so that the values were in descending order. We then generated a final outcome where the distribution was dependent on the weights and the confounders.

The true causal effect within our synthetic data will be the weight of the higher rank minus the weight for the smaller rank, which will result in a negative coefficient that models the decreasing probability of suspension as rank increases. We ran our backdoor estimator on the synthetic data and extracted the coefficient of the treatment, which represents our observed causal effect. We compared this output to the result of a logistic regression that only considered the treatment, which gave us a naive estimate $E[Y|A]$, which represents the baseline that we compared our estimator to.

After running our model on sample datasets with sizes 1,000, 10,000, and 100,000, our results are listed below:

Sample Size	Expected Causal Effect	Observed Causal Effect	Observed % Error	Naive Estimate	Naive % Error
1,000	-0.144	-0.179	0.243	-0.101	0.299
10,000	-0.168	-0.166	0.0119	-0.0958	0.430
100,000	-0.151	-0.149	0.0132	-0.0771	0.541

At every level, our backdoor estimator outperforms the naive estimate, and this difference is especially clear at the larger sample sizes. While our backdoor estimator hovers around an error of 1%, the naive estimation error ranges from 43-54%, which shows that our model performs remarkably well compared to the base model. This means that our estimator is set up effectively and is able to discern the effects of the treatment while controlling for the impact of the confounders.

7. Reflections

What was interesting?

The main benefit of this project is that we had apply the concepts we had learned in class to build our own model on real data, which is an invaluable experience that pushed us to develop a deeper understanding of how causal inference is implemented in the real world, and all the issues

that make it such a complex topic. Most of our project relied on lectures from the first half of the quarter, especially since we were focused on implementing a backdoor estimator on our causal DAG. We relied on many of the methods that we wrote for HW2 in our analysis, especially for backdoor, IPW, and bootstrap. For the final report, we also wanted to explore using DoWhy and DoubleML to incorporate more flexibility and complexity into our models, and be able to handle more high dimensional confounders.

As there is anecdotal evidence suggesting that police officers are not always accurately punished for their actions, we were interested in exploring this through a quantitative analysis. This project highlighted the importance of data analysis in uncovering underlying patterns within institutional structures. Additionally, working with real-world data presented the opportunity to uncover patterns and trends that are not immediately visible through anecdotal evidence alone. The use of advanced causal inference techniques, such as double machine learning and the construction of a detailed DAG, provided a deeper understanding of the complexities within the data and relationships between different variables.

What was difficult?

A lot of the initial difficulty came from processing the extremely large and disjoint datasets that were available to us through Github. We had to spend a lot of time figuring out what data we actually wanted to grab and use, what question we wanted to answer, and if we had enough data to isolate the identified causal effect for our question—at times, it really felt like a chicken and egg dilemma. Handling the multiple confounders and their interactions also required careful consideration. For another group starting the same project, I'd recommend that they try to get a causal DAG down as early as possible after an initial assessment of the data. We went through a lot of refinement from our first model to now, but having a graph to base our model on and iterate off of was really helpful for us to actually start creating estimates and figuring out what other confounders we wanted to include.

Unaddressed Challenges

There are a lot of unknown confounders in our data. Including an officer's prior disciplinary history, specific details about allegations, and duties by rank would improve our model. We also lack complainant information, such as race and whether the complainant is an officer or civilian, which could significantly impact the causal effect. This makes precise estimation challenging. Additionally, our analysis is based on a relatively small sample from a single police force, limiting the generalizability of our findings. Unaccounted departmental policies and non-representative data further complicate drawing broader conclusions.

With the bootstrapping and IPW analyses, we think that the smaller sample size is less of the issue compared to the unknown confounders. There are just so many possible variables in a human interaction that could play into the final outcome that it's very difficult to say that we've

truly isolated the effect of one treatment. This is an example where our assumptions of consistency are probably just not realistic: there's no way that we can cleanly intervene on rank in a way that doesn't have potential ripple effects on other variables we can't control for.

What's left to do?

If we had more time and computing resources, we'd want to expand out our DoubleML model and include more categorical variables in our confounders. It'd be fun to experiment with a variety of different models and see which one is more suitable for our data. We also didn't have time to fully build out another model with final finding as an alternate outcome, since the IU article we cited in the update brought to our attention the fact that final finding might also include racial discrepancies.

Also, we would like to extend our project to also include a qualitative analysis with more . With additional time and monetary resources, we would gather more detailed data, including interviews, to capture nuances such as departmental politics not reflected in the quantitative data. We would also explore and implement more sophisticated modeling techniques, such as hierarchical models or neuron network models, to better account for the structure of the data.

With all of this, we believe that we could make a stronger causal conclusion, and potentially paint a clearer picture of whether the military euphemism "different spans for different ranks" also perpetuates within our police system.